INCH-POUND

MIL-S-82858(OS)
29 December 1989
SUPERSEDING
WS 26951
13 November 1986

MILITARY SPECIFICATION

SILICONE DIOXIDE, MICROFINE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of microfine silicone dioxide, referred to herein as silicone dioxide.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

FEDERAL

FED-STD-313 Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 9620

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 11 Wire-Cloth Sieves for Testing Purposes
(DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample of the silicone dioxide shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Chemical and physical properties. Chemical and physical properties of the silicone dioxide shall conform to table I.

TABLE I. Chemical and physical properties.

Property	Limits	
	Minimum	Maximum
Moisture, percent (at 105°C)	---	2.0
Loss on ignition, percent (dry basis)	---	1.5
pH (4 percent aqueous dispersion	3.5	6.0
Bulk density, grams/cc	0.03	0.04

3.3 Safety.

3.3.1 Toxic products and formulations. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

3.3.2 Material Safety Data Sheets (MSDS). The contractor shall prepare and submit an MSDS in accordance with FED-STD-313 as specified in the contract (see 6.2).

3.4 Shelf life. The silicone dioxide specified herein shall have an unlimited shelf life when stored at ambient conditions (see 6.3.1).

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3.5 **Workmanship.** Workmanship shall be such that the silicone dioxide is uniform in appearance and free from visible contamination.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 **Toxicological product formulations.** The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 **First article inspection.** First article inspection shall consist of the quality conformance inspection (see 4.4). Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

4.4 **Quality conformance inspection.** Quality conformance inspection shall consist of the tests specified in table II.

4.4.1 **Lot.** A lot shall consist of all silicone dioxide submitted for inspection at one time from the same production run.

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TABLE II. Quality conformance inspection.

Examinations and Tests	Requirement	Test Method
Visual examination	3.5 and Section 5	4.5.1
Moisture	3.2	4.5.2
Loss on ignition	3.2	4.5.3
pH	3.2	4.5.4
Bulk density	3.2	4.5.5

4.4.2 **Sampling.** Sampling shall be in accordance with table III.

TABLE III. Sampling.

Number of containers in lot	Number of containers to be selected
1 - 3	all
4 - 40	3
41 - 65	4
66 - 110	5
111 - 500	7

4.5 **Inspection methods.** The following inspection methods shall be used. Unless otherwise specified in the applicable test method; all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.5.1 **Visual examination.** Visual examination shall be made to verify conformance to the requirements of 3.5 and section 5.

4.5.2 **Moisture.** Moisture shall be determined in accordance with the following:

a. Place a 2- to 3-gram (g) sample of silicone dioxide, weighed to the nearest 0.1 milligram (mg), into a dry, tared 100-milliliter (ml) crucible.

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b. Place the crucible and sample into a 105 ± 5°C oven for 60 ± 5 minutes with the cover off the crucible.

c. Replace the cover and remove the sample to a desiccator. Cool and weigh the sample.

d. Calculation:

$$\text{Percent moisture} = \frac{(B - C)}{(B - A)} \times 100$$

Where: A = weight of tared crucible and cover

B = weight of tared crucible, cover, and sample

C = weight of tared crucible, cover, and dried sample

e. Record moisture to nearest 0.1 percent.

4.5.3 Loss on ignition. Loss on ignition shall be determined in accordance with the following:

a. Into a previously heated and tared 100-ml crucible, weigh to the nearest 0.1 mg a 2- to 3-g dried sample of silicone dioxide that was used in moisture determination.

b. Place the sample into a cool muffle furnace and adjust the crucible cover so any material volatized may escape.

c. Bring the temperature to 1000 ± 50°C and allow the sample to remain at this temperature for 15 +5 -0 minutes.

d. Readjust the cover to completely cover the crucible and place the crucible and sample into a desiccator. Cool and weigh the sample.

e. Calculation: $$\text{Percent ignition loss} = \frac{(B - C)}{(B - A)} \times 100$$

Where: A = weight of tared crucible and cover

B = weight of tared crucible, cover, and sample

C = weight of tared crucible, cover, and ignited sample

f. Record loss on ignition to the nearest 0.1 percent.

4.5.4 pH. The pH shall be determined in accordance with the following:

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a. Weigh 4.0 ± 0.5 g of silicone dioxide into a 250-ml beaker.

b. Add 100 ml of distilled water that has been adjusted to a pH of 7.0 with 0.01 normal (N) hydrochloric acid or sodium hydroxide.

c. Stir the suspension mechanically for 5 minutes and determine the pH using a pH meter with a glass-calomel electrode pair.

d. Record the pH to the nearest 0.1 pH unit.

4.5.5 Bulk density. Bulk density shall be determined in accordance with the following:

a. Weigh a clean, dry, 100-ml calibrated glass cylinder to the nearest 0.01 g.

b. Support a 7.5 ± 1.5-cm powder funnel with the discharge opening centered 3 cm above the top of the cylinder.

c. Pass 120 ± 24 ml of the sample through a No. 10 sieve conforming to ASTM E 11 onto glazed paper. Examine any lumps retained on the screen for conformance to 3.5

NOTE: Soft lumps that can be readily dispersed are not cause for rejection.

d. Slowly shake the sifted sample from the paper into the funnel.

e. Strike off the excess sample from the top of the cylinder.

f. Wipe the exterior of the cylinder, and weigh the sample and cylinder.

g. Calculation: Bulk density, g/cc = $\frac{(A - B)}{V}$

Where: A = weight of cylinder and sample, g

B = weight of cylinder, g

V = volume of calibrated cylinder, ml

h. Record the bulk density to the nearest 0.01 g/cc.

4.6 Inspection of packaging. The containers and container markings shall be visually examined to verify compliance with section 5.

5. PACKAGING

5.1 Packaging. Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

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5.1.1 **Level C.** The silicone dioxide shall be packaged to afford adequate protection against loss, contamination, deterioration, and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.4. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking.** In addition to the marking requirements of 5.2.1, each container marking shall include the following:

a. Title, number, and date of this specification.

b. Manufacturer's name and address.

c. Date of manufacture.

d. Net weight of contents.

e. Shelf life.

f. Storage conditions.

g. Toxicity warning (if applicable).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 **Intended use.** The silicone dioxide is intended for use as a filler in preparing insulations and liners in the manufacture of Mk 104 Standard Missile Dual Thrust Rocket Motors.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).

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c. Whether first article inspection is required (see 3.1 and 6.4).

d. Material Safety Data Sheets Required (see 3.3.2 and 6.6).

e. Special marking, if other than as specified in 5.2.2.

6.3 **Definition.**

6.3.1 **Ambient conditions.** Ambient conditions are those conditions of temperature and relative humidity experienced by the material when stored in an enclosed warehouse.

6.4 **First article.** When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 **Possible material.** The following material has been found to meet the requirements of this specification. This is only given for information and is not restrictive. Cab-o-Sil M-5 manufactured by Cabot Corporation, 125 High Street, Boston, Mass. 02110, has been found to meet the requirements of this specification.

6.6 **Material Safety Data Sheets.** MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) Part 23, subpart 23.3. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.7 **Subject term (key word) listing.**

Rocket Motor, Dual Thrust, MK 104
Standard Missile

Preparing Activity:
NAVY-OS
(Project 9620-N015)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-82858	2. DOCUMENT DATE (YYMMDD) 29 December 1989

3. DOCUMENT TITLE
SILICONE DIOXIDE, MICROFINE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	e. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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